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ICS 2203 WEB APPLICATION DEVELOPMENT 1

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ICS 2203 WEB APPLICATION DEVELOPMENT 1

This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 7

XHTML

Learning outcomes

Upon completing this topic, you should be able to:

1. Identify XHTML web technologies used in web programming.
2. Employ XHTML technology in creating simple web pages.
3. Generate web pages though the use XHTML.

7.1. Introduction

One of the major changes to HTML which was introduced to XHTML is that tags must always be properly formed. With



ICS 2203 WEB APPLICATION DEVELOPMENT 1

the old HTML specification you could be very sloppy in your coding, with missing tags and incorrect formation without many problems but in XHTML this is very important. XHTML is defined as a stricter version of HTML with the enforcement of syntax and rules of the language. XHTML was created for two main reasons:

- To create a stricter standard for making web pages, reducing incompatibilities between browsers –
- To create a standard that can be used on a variety of different devices without changes

These rules define the use of lower case, nesting of tags and elements, addition of attributes and attribute values and the closing of tags.



7.2. Lower Case

In XHTML, tags must always be lower case. This means that:

`<FONT>`

`<Font>`

`<FoNT>`

are all incorrect tags and must not be used. The font tag must now be used as follows:

`<font>`

If you are not writing your code, but instead use a WYSIWYG editor, you can still begin to migrate your documents to XHTML by setting the editor to output all code in lower case. For example, in Dreamweaver 4 you can do this by going to:

Edit -> Preferences -> Code Format

and making sure that Case For Tags is set to:



<lowercase>

and also that Case For Attributes is set to:

lowercase="value"

7.3. Nesting

In XHTML tags must be properly nested. This means that if you have multiple tags applying to something on your page you must make sure you open and close them in the correct order. For example if you have some bold red text in a paragraph, the correct nesting would be one of the following:

```
<p><b><font color=
"#FF0000">Your Text</font></b></p>
```

```
<b><p><font color=
"#FF0000">Your Text</font></p></b>
```



```
<p><font color=
```

```
"#FF0000"><b>Your Text<b></font></p>
```

These are only examples, though, and there are other possibilities for these tags. What you must not do, though, is to close tags in the wrong order, for example:

```
<p><b><font color=
```

```
"#FF0000">Your Text</p></font></b>
```

Although code in this form would be shown correctly using HTML, this is incorrect in the XHTML specification and you must be very careful to nest your tags correctly.

7.4. Closing Tags

The previous two changes to HTML should not be a particular problem to you if your HTML code is already well formed. The



ICS 2203 WEB APPLICATION DEVELOPMENT 1

final change to HTML tags probably will require quite a lot of changes to your HTML documents to make them XHTML compliant.

All tags in XHTML must be closed. Most tags in HTML are already closed (for example `<p></p>`, `<font></font>`, `<b></b>`) but there are several which are standalone tags which do not get closed. The main three are:

`<br>`

`<img>`

`<hr>`

There are two ways in which you can deal with the change in specification. The first way is quite obvious if you know HTML. You can just add a closing tag to each one, e.g.

`<br></br>`



`<img></img>`

`<hr></hr>`

Although you must be careful that you do not accidentally place anything between the opening and closing tags as this would be incorrect coding. The second way is slightly different but will be familiar to anyone who has written WML. You can include the closing in the actual tag:

`<br />`

`<img />`

`<hr />`

This is probably the best way to close your tags, as it is the recommended way by the W3C who set the XHTML standard. You should notice that, in these examples, there is a space before the `/>`. This is not actually necessary in the XHTML



ICS 2203 WEB APPLICATION DEVELOPMENT 1

specification (you could have `<br/>`) but the reason why I have included it is that, as well as being correct XHTML, it will also make the tag compatible with past browsers. As every other XHTML change is backwards compatible, it would not be very good to have a simple missed out space causing problems with site compatibility.

In case you are wondering how the `<img>` tag works if it has all the normal attributes included, here is an example:

```

```

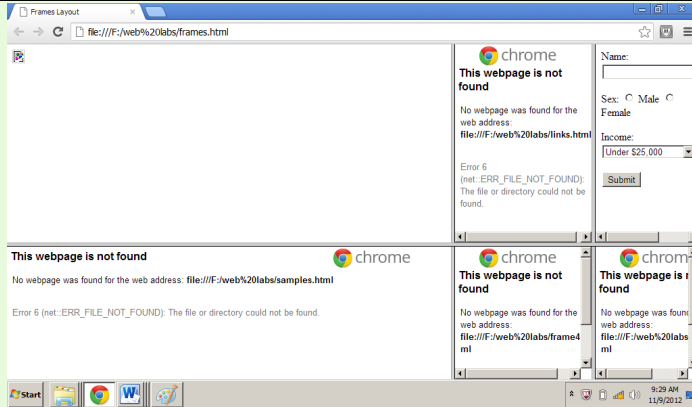
Again, notice the space before the `/>`

7.4.1. Consider the browser output

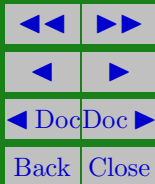
EXERCISE 1. 



ICS 2203 WEB APPLICATION DEVELOPMENT 1



Required: write xhtml that will provide the output shown above





7.4.2. The Doctype

- The first change which will appear on your page is the Doctype. When using HTML it is considered good practice to add a Doctype to the beginning of the page like this.
- All though this was optional in HTML, XHTML requires you to add a Doctype. There are three available for use.

Strict - This is used mainly when the markup is very clean and there is no 'extra' markup to aid the presentation of the document. This is best used if you are using Cascading Style Sheets for presentation.

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0
```

Strict



ICS 2203 WEB APPLICATION DEVELOPMENT 1

```
//EN" "http://www.w3.org/TR/xhtml1  
/DTD/xhtml1-strict.dtd">
```

- **Strict**

This is used mainly when the markup is very clean and there is no 'extra' markup to aid the presentation of the document. This is best used if you are using Cascading Style Sheets for presentation.

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0
```

Strict

```
//EN" "http://www.w3.org/TR/xhtml1  
/DTD/xhtml1-strict.dtd">
```



ICS 2203 WEB APPLICATION DEVELOPMENT 1

- **Transitional**

This should be used if you want to use presentational features of HTML in your page.

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0
```

Transitional

```
//EN" "http://www.w3.org/TR/xhtml1  
/DTD/xhtml1-transitional.dtd">
```

- **Frameset**

This should be used if you want to have frames on your page.

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0
```

Frameset

```
//EN" "http://www.w3.org/TR/xhtml1  
/DTD/xhtml1-frameset.dtd">
```



The doctype should be the very first line of your document and should be the only thing on that line. You don't need to worry about this confusing older browsers because the Doctype is actually a comment tag.

7.4.3. The XHTML Document Structure

After the Doctype line, the actual XHTML content can be placed. Similar to HTML, XHTML has `<html>` `<head>` `<title>` and `<body>` tags but, unlike with HTML, they must all be included in a valid XHTML document. The correct setup of your file is as follows:

Requirements

Similar to HTML standard XHTML documents Source-code is created in a Text editor (e.g. Notepad, Wordpad, emacs, etc.)



ICS 2203 WEB APPLICATION DEVELOPMENT 1

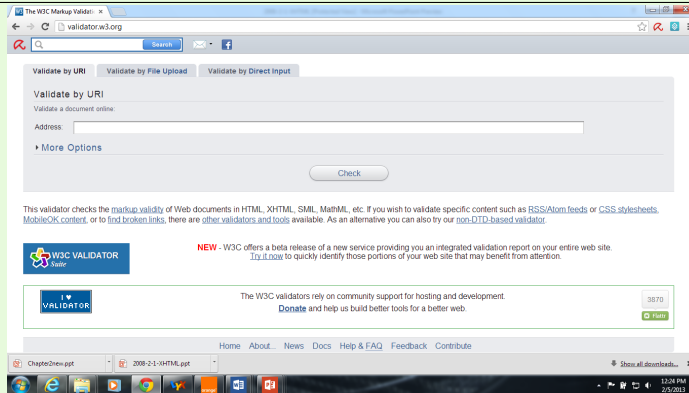
and will have the file name extensions .html or .htm file-name
.Web server Stores XHTML documents Web browser Requests
XHTML documents using the normal HTTP request from a
client.

The W3C offers a validation service whereby after creating
your xhtml document you may submit it to their server and
have it validated to see if it conforms to the set standards and
guidelines. the VALidator service checks the document syntax
and may be found at (validator.w3.org/file-upload.html)

ICS 2203 WEB APPLICATION DEVELOPMENT 1

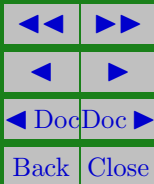


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Example . Example of an XHTML transitional document-Document: Type the code below a note pad Editor and save it as validation example.html. Submit it to the W3C validator site

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-
```





ICS 2203 WEB APPLICATION DEVELOPMENT 1

transitional.dtd">

<html> <body>

preceding content...

 <h3>Kumquat Stew Recipes</h3>

<i>In the same or another</i>

For more information on kumquats, visit our

Kumquat Archive,

and perhaps try one or two of our

Kumquat Stew Recipes.

<p>



ICS 2203 WEB APPLICATION DEVELOPMENT 1

```
<li><a href="default.html">Home</a></li>
```

```
<li><a href="about/about.htm">About</a></li>
```

```
<li><a href="news/news.htm">News</a></li>
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0
```

Frameset

```
//EN" "http://www.w3.org/TR/xhtml1
```

```
/DTD/xhtml1-frameset.dtd">
```

```
<li><a href="calendar/calendar.htm">Calendar</a></li>
```

```
<li><a href="photo_gallery/photo_gallery.htm">
```

```
Photo Gallery</a></li>
```

```
<li><a
```

```
href="information_links/information_links.htm">Links</a>
```

```
</li> <li><a href="faq/faq.htm">FAQ</a></li>
```

```
<li><a href="contact/contact.htm">Contact</a></li>
```



<table border cellpadding=0 cellspacing=5>

<caption align=bottom>

Kumquat versus a poked eye, by gender</caption>

<tr>

<td colspan=2 rowspan=2></td>

<th colspan=2 align=center>Preference</th>

</tr>

<tr>

<th>Eating Kumquats</th>

<th>Poke In The Eye</th>

</tr>

<tr align=center>

<th rowspan=2>Gender</th>



ICS 2203 WEB APPLICATION DEVELOPMENT 1

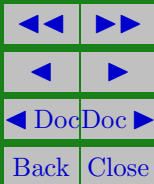
```
<th>Male</th>
<td>73%</td>
<td>27%</td>
</tr>
<tr align=center>
<th>Female</th>
<td>16%</td>
<td>84%</td>
</tr>
</table>
</body>
</html>
```

Solution: Put answer





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Revision Questions

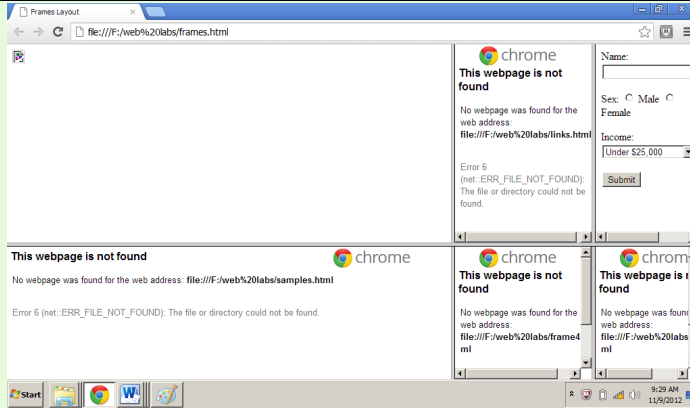
EXERCISE 2. 🖱️ Illustrate any four benefits of XHTML.

EXERCISE 3. 🖱️ Identify the three common DOC types of an XHTML document.

EXERCISE 4. 🖱️ Consider the browser output



ICS 2203 WEB APPLICATION DEVELOPMENT 1



Required: write xhtml that will provide the output shown above

EXERCISE 5. ✍ Differentiate between HTML and XHTML

EXERCISE 6. ✍ When do we use XHTML instead of HTML





References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp
- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5



Solutions to Exercises

Exercise 1. `<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML`

1.0 Frameset

```
//EN" "http://www.w3.org/TR/xhtml1/DTD
/xhtml1-frameset.dtd">
<html>
<head>
<title>Frames Layout</title>
</head>
<frameset rows="60%,*" cols="65%,20%,*">
<frame src="image.html">
<framesrc="links.html">
<frame src="form.html">
<frame src="samples.html" name="fill_me">
```



```
<frame scrolling=yes src="frame4.html">
```

```
<frame src="frame5.html">
```

```
<frame src="frame6.html">
```

```
<noframes>
```

Sorry, this document can be viewed only with a frames-capable browser.

```
<a href = "frame1.html">Take this link</a>
```

to the first HTML document in the set.

```
</noframes>
```

```
</frameset>
```

```
</html>
```

Exercise 1





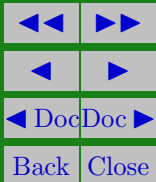
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Exercise 2.

Put answer

Exercise 2





Exercise 3.

The Transitional DOC TYPE

The Frameset DOC TYPE

The Strict DOC TYPE

Exercise 3



Exercise 4.

```
<!DOCTYPE html PUBLIC "-
//W3C//DTD XHTML 1.0 Frameset
//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.d
<html>
<head>
<title>Frames Layout</title>
</head>
<frameset rows="60%," cols="65%,20%,">
<frame src="image.html">
<framesrc="links.html">
<frame src="form.html">
<frame src="samples.html" name="fill_me">
<frame scrolling=yes src="frame4.html">
```



ICS 2203 WEB APPLICATION DEVELOPMENT 1

```
<frame src="frame5.html">
```

```
<frame src="frame6.html">
```

```
<noframes>
```

Sorry, this document can be viewed only with a frames-capable browser.

```
<a href = "frame1.html">Take this link</a>
```

to the first HTML document in the set.

```
</noframes>
```

```
</frameset>
```

```
</html>
```

Exercise 4



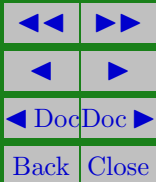
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Exercise 5.

Put answer

Exercise 5





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Exercise 6.

Put answer

Exercise 6

